MIL-F-60348B (MU)
1 February 1970
SUPERSEDING
MIL-F-60348A (MU)
16 August 1968

MILITARY SPECIFICATION

FUZE, POINT DETONATING, M48A3, METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for the M48A3 point detonating artillery ammunition fuze.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1390

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DRAWINGS

ARMY

8798222 - Body Assembly
8798228 - Setting Sleeve Assembly
8798235 - Head Assembly, Metal Parts
9234456 - Bottom Closing Screw
9234453 - Fuze, PD, M48A3 Assembly Parts Arrangement

PUBLICATIONS

ARMY

ET-9234453 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-The materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawing (dwg.) 9234453, and with all requirements specified in applicable specifications.

3.3 Spring embrittlement.-Tension and interrupter springs shall not break when extended to the minimum (min.) point of permanent distortion when tested as specified in 4.4.1.

3.4 Torque of setting sleeve on body assembly.-The torque required to rotate the setting sleeve between the "superquick" and "delay" positions shall be not less than 5 inch-ounces nor more than 48 inch-ounces when tested as specified in 4.4.3.

3.5 Workmanship.-All parts shall be fabricated and finished in a thorough, workmanlike manner and all manufacturing, processing and assembly operations shall be correctly performed. The parts shall be clean and free of burrs, sharp edges, unblended radii, surface defects, cracks, chips, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible, and sharply defined.

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3.6 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual ducuments, and the certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer (see 6.4)

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for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 25 complete sets of parts (i.e., 25 each of every component and every subassembly) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures, and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component to comply with any of the stated requirements.

4.3 Inspection Provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for

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submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Support, firing pin (see dwg. 8798238, covering a detail of dwg. 8798235).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Concentricity of outside diameter with hole. (NOTE: Will be inspected as the first article sample only, and as a first article sample each time the tooling is changed)	Gage	01001
102.	Crack	Visual	01002
Minor:	AQL 2.5 Percent		
201.	Outside diameter	Gage	01003
202.	Inside diameter	Gage	01004

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203.	Diameter of hole	Gage	01005
204.	Thickness of bottom	Gage	01006
205.	Overall length	Gage	01007
206.	Evidence of poor workmanship (see 3.5)	Visual	01008

4.3.2.2 Head (see dwg. 8798237, covering a detail of dwg. 8798235).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 Percent		
101.	Minor diameter of flash tube thread, max.	Gage	02001
102.	Pitch diameter of flash tube thread	Gage	02002
103.	Depth of flash tube thread, min..	Gage	02003
104.	Pitch diameter of detonator retaining screw thread	Gage	02004
105.	Diameter of firing pin assembly cavity	Gage	02005
106.	Diameter of firing pin hole	Gage	02006
107.	Concentricity of flash tube hole thread with firing pin hole.	Gage	02007
Minor:	AQL 4.0 Percent		
201.	Depth of flash tube cavity	Gage	02008
202.	Depth to bottom of detonator retaining screw thread, min.	Gage	02009
203.	Diameter of detonator cavity	Gage	02010
204.	Diameter of support cup counterbore, max.	Gage	02011
205.	Distance to ogive seat flange, min.	Gage	02012
206.	Outside diameter of flange, min.	Gage	02013
207.	Depth to bottom of firing pin assembly cavity	Gage	02014
208.	Outside diameter under flange ...	Gage	02015
209.	Total length	Gage	02016
210.	Outside diameter at nose	Gage	02017
211.	Concentricity of detonator cavity with firing pin hole	Gage	02018
212.	Minor diameter of detonator retaining screw thread	Gage	02019
213.	Wrench hole or ogive staking hole missing	Visual	02020
214.	Evidence of poor workmanship (see 3.5)	Visual	02021

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4.3.2.3 Pin, firing (see dwg. 9234572 covering a detail of dwg. 8798235).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 Percent		
101.	Distance from point to base	Gage	03001
102.	Diameter and profile of pin point	Gage	03002
103.	Thickness of base	Gage	04001
104.	Concentricity of base with firing pin	Gage	04002
105.	Evidence of poor workmanship (see 3.5)	Visual	03004
Minor:	AQL 1.5 Percent		
201.	Diameter of pin shaft	Gage	03003
202.	Diameter of base	Gage	04004
203.	Diameter of base at pin union ...	Gage	03005

4.3.2.4 Washer, closing disc (see dwg. 8798239, covering a detail of dwg. 8798235).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined		
Minor:	AQL 2.5 Percent		
201.	Distance between flats	Gage	05001
202.	Thickness	Gage	05002
203.	Outside diameter	Gage	05003
204.	Inside diameter	Gage	05004
205.	Evidence of poor workmanship (see 3.5)	Visual	05005

4.3.2.5 Disc, closing (see dwg. 8798236, covering a detail of dwg. 8798235).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Crack or perforation	Visual	06001

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Minor:	AQL 1.5 Percent		
201.	Thickness	Gage	06002
202.	Diameter	Gage	06003
203.	Evidence of poor workmanship (see 3.5)	Visual	06004

4.3.2.6 Head, metal parts assembly (see dwg. 8798235, covering a detail of dwg. 9234453).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Distance from firing pin point to detonator cavity, min.	Gage	07001
Major:	AQL 0.40 Percent		
101.	Distance from firing pin point to detonator cavity, max. .	Gage	07002
102.	Any component missing, damaged, or improperly assembled	Visual	07003
Minor:	AQL 1.0 Percent		
201.	Closing disc crimp (less than 360 degrees, or disc loose)	Visual-Manual	07004
202.	Evidence of poor workmanship (see 3.5)	Visual	07005

4.3.2.7 Tube, flash (see dwg. 8798221, covering a detail of dwg. 9234453).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.65 Percent		
101.	Pitch diameter of threads	Gage	08001
102.	Length of threads, min.	Gage	08002
103.	Evidence of poor workmanship, especially burrs or chips within flash tube	Visual	08003
Minor:	AQL 2.5 Percent		
201.	Major diameter of threads, max. .	Gage	08004
202.	Total length	Gage	08005
203.	Length of thread pilots, max. ...	Gage	08006
204.	Diameter of thread pilots	Gage	08007

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205.	Diameter of flash hole, max.	Gage	08008
206.	Plug gage fails to pass through flash hole	Gage	08009

4.3.2.8 Ogive (see dwg. 8798220, covering a detail of dwg. 9234453).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 Percent		
101.	Location of setting sleeve hole	Gage	09001
102.	Diameter of setting sleeve hole	Gage	09002
103.	Wall thickness	Gage	09003
104.	Inside diameter of head end	Gage	09004
105.	Through crack	Visual	09005
Minor:	AQL 4.0 Percent		
201.	Inside diameter at body end	Gage	09006
202.	Total length	Gage	09007
203.	Outside diameter of body end, min	Gage	09009
204.	Location of inside projections ..	Gage	09010
205.	Distance between inside projections	Gage	09011
206.	Distance from centerline to inside projections	Gage	09012
207.	Marking incorrect, unidentifiable or incorrectly positioned	Visual	09013
208.	Evidence of poor workmanship, cracks or dents (see 3.5)	Visual	09014
209.	Protective finish missing, damaged, or contains bare spots .	Visual	09015

4.3.2.9 Interrupter (see dwg. 8798224, covering a detail of dwg. 8798222).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Diameter, min.	Gage	10001
Major:	AQL 0.65 Percent		
101.	Diameter, max.	Gage	10002
102.	Total length	Gage	10003
103.	Improper finish	Visual	10004

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Minor:	AQL 1.0 Percent		
201.	Protective finish missing or contains bare spot	Visual	10005
202.	Evidence of poor workmanship (see 3.5)	Visual	10006

4.3.2.10 Spring, tension (see dwg. 8798225, covering a detail of dwg. 8798222).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 2.5 Percent		
201.	Outside diameter, max.	Gage	11001
202.	Inside diameter, min.	Gage	11002
203.	Wire diameter	Gage	11003
204.	Protective finish, missing or damaged	Visual	11004
205.	Evidence of poor workmanship (see 3.5)	Visual	11005

4.3.2.11 Retainer setting sleeve (see dwg. 8798226, covering a detail of dwg. 8798222).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Pitch diameter of thread, min. ..	Gage	12001
Minor:	AQL 4.0 Percent		
201.	Diameter of thread pilot	Gage	12002
202.	Diameter of hole	Gage	12003
203.	Diameter of hole counterbore	Gage	12004
204.	Depth of hole counterbore .....Gage		12005
205.	Length	Gage	12006
206.	Major diameter of thread, min. ..	Gage	12007
207.	Length of thread pilot	Gage	12008
208.	Width of screwdriver slot	Gage	12009
209.	Protective finish missing or damaged	Visual	12010
210.	Evidence of poor workmanship (see 3.5)	Visual	12011

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4.3.2.12 Washer, setting sleeve (see dwg. 8798227, covering a detail of dwg. 8798222).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major: None defined.			
Minor:	AQL 2.5 Percent		
201.	Thickness	Gage	13001
202.	Outside diameter	Gage	13002
203.	Inside diameter	Gage	13003
204.	Improper finish	Visual	13004
205.	Evidence of poor workmanship (see 3.5)	Visual	13005
206.	Protective finish missing or damaged	Visual	13006

4.3.2.13 Sleeve, setting (see dwg. 8798229, covering a detail of dwg. 8798228).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 1.0 Percent		
101.	Location of spring cup and spring cavity	Gage	14001
102.	Depth of spring cup cavity	Gage	14002
103.	Depth of spring cavity	Gage	14003
104.	Diameter of spring cup cavity	Gage	14004
105.	Distance from base to shoulder	Gage	14005
106.	Outside diameter	Gage	14006
107.	Location of setting slot	Gage	14007
108.	Width of setting slot	Gage	14008
109.	Spring cup cavity finish improper	Visual	14009
Minor:	AQL 2.5 Percent		
201.	Total length	Gage	14010
202.	Diameter of neck	Gage	14011
203.	Depth of slot	Gage	14012
204.	Diameter of spring cup cavity counterbore	Gage	14013
205.	Depth of spring cup cavity counterbore	Gage	14014

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206.	Diameter of spring cavity	Gage	14015
207.	Evidence of poor workmanship (see 3.5)	Visual	14016
208.	Protective finish missing or contains bare spot	Visual	14017

4.3.2.14 Cup, spring (see dwg 8798230, covering a detail of dwg. 8798228).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Large outside diameter, max.	Gage	15001
102.	Outside diameter finish improper .	Visual	15002
103.	Hole missing or obstructed	Visual	15003
Minor:	AQL 2.5 Percent		
201.	Large outside diameter, min.	Gage	15004
202.	Depth of spring cavity	Gage	15005
203.	Diameter of spring cavity	Gage	15006
204.	Length	Gage	15007
205.	Small outside diameter	Gage	15008
206.	Protective finish missing or contains bare spot	Visual	15009
207.	Evidence of poor workmanship (see 3.5)	Visual	15010

4.3.2.15 Spring, interrupter (see dwg. 8798231, covering a detail of dwg. 8798228).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Solid height	Gage	16001
102.	Outside diameter, max.	Gage	16002
Minor:	AQL 1.5 Percent		
201.	Inside diameter, min.	Gage	16003
202.	Ends not closed	Visual	16004
203.	Evidence of poor workmanship (see 3.5)	Visual	16005
204.	Protective finish missing or damaged	Visual	16006

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4.3.2.16 Setting sleeve assembly, when applicable (see dwg. 8798228, covering a detail of dwg. 8798222).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Any component missing, damaged, or improperly assembled	Visual-Manual	17001
Minor:	AQL 1.0 Percent		
201.	Distance from bottom of spring cup to end of setting sleeve, max.	Gage	17002
202.	Evidence of poor workmanship (see 3.5)	Visual	17003

4.3.2.17 Body (see dwg. 8798223, covering a detail of dwg. 8798222).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Dia. of interrupter cavity, max. .	Gage	18001
Major:	AQL 1.5 Percent		
101.	Pitch diameter of external thread	Gage	18002
102.	Major diameter of external thread	Gage	18003
103.	Minor diameter bottom closing screw thread, max.	Gage	18004
104.	Pitch diameter bottom closing screw thread	Gage	18005
105.	Minor diameter flash tube thread, max.	Gage	18006
106.	Pitch diameter of flash tube thread	Gage	18007
107.	Symmetry of interrupter cavity with flash hole	Gage	18008
108.	Diameter under flange	Gage	18009
109.	Pitch diameter of retainer sleeve thread	Gage	18010
110.	Distance interrupter cavity protrudes beyond centerline	Gage	18011
111.	Concentricity of pitch diameter of flash tube hole with flash hole	Gage	18012

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112.	Depth from base to bottom of delay plunger cavity	Gage	18013
113.	Diameter of interrupter cavity, min.	Gage	18014

Minor: AQL 6.5 Percent

201.	Thickness of boss	Gage	18015
202.	Thickness of flange	Gage	18016
203.	Distance between wrench slots	Gage	18017
204.	Width of wrench slots, min.	Gage	18018
205.	Location of setting sleeve cavity.	Gage	18019
206.	Location of interrupter cavity ...	Gage	18020
207.	Total length	Gage	18021
208.	Depth of flash tube hole	Gage	18022
209.	Distance from base end to flange .	Gage	18023
210.	Distance from bottom of retainer sleeve hole to thread, max.	Gage	18024
211.	Length of external thread undercut, max.	Gage	18025
212.	Length of external thread pilot, max.	Gage	18026
213.	Length of bottom closing screw thread, min.	Gage	18027
214.	Depth of delay plunger cavity from shoulder	Gage	18028
215.	Diameter of delay plunger cavity	Gage	18029
216.	Concentricity of flash hole with delay plunger cavity	Gage	18030
217.	Minor diameter retainer sleeve thread, max.	Gage	18031
218.	Depth of interrupter cavity	Gage	18032
219.	Depth of setting sleeve cavity ...	Gage	18033
220.	Distance from bottom of flash tube hole to thread, max.	Gage	18034
221.	Diameter of setting sleeve cavity	Gage	18035
222.	Diameter of flange, min.	Gage	18036
223.	Angle of boss	Gage	18037
224.	Diameter of external thread undercut, min.	Gage	18038
225.	Diameter of flash hole	Gage	18039
226.	Diameter of ogive seat	Gage	18040
227.	Small diameter of flange	Gage	18041
228.	Distance from base to bottom of external thread undercut	Gage	18042
229.	Diameter of external thread pilot.	Gage	18043
230.	Concentricity of external thread with flash hole	Gage	18044

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231.	Length of flange	Gage	18045
232.	Diameter of flash tube hole undercut, min.	Gage	18046
233.	Distance from edge of boss to centerline of interrupter cavity, min.	Gage	18047
234.	Thickness of flange at wrench slot	Gage	18048
235.	Protective finish missing or contains bare spots	Visual	18049
236.	Existence of flash hole countersink	Visual	18050
237.	Evidence of poor workmanship (see 3.5)	Visual	18051

4.3.2.18 Body assembly (see dwg. 8798222, covering a detail of dwg. 9234453).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Interrupter in armed or partly armed position	Visual	19001
Major:	AQL 0.65 Percent		
101.	Thread damaged	Visual	19002
102.	Setting sleeve slot incorrectly positioned	Visual	19003
103.	Retainer setting sleeve stake missing	Visual	19004
Minor:	AQL 1.0 Percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	19005

4.3.2.19 Screw, bottom closing (see dwg. 9234456).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.65 Percent		
101.	Major diameter of thread	Gage	21001
102.	Pitch diameter of thread	Gage	21002
103.	Concentricity of delay plunger cavity with threads	Gage	21003
Minor:	AQL 4.0 Percent		
201.	Diameter of delay plunger cavity .	Gage	21007
202.	Diameter of thru-hole	Gage	21008
203.	Diameter of wrench holes, min. ...	Gage	21009

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204.	Depth of delay plunger cavity	Gage	21010
205.	Distance between wrench holes	Gage	21011
206.	Total length	Gage	21013
207.	Concentricity of thru-hole with threads	Gage	21014
208.	Evidence of poor workmanship (see 3.5)	Visual	21015

4.3.3 Testing

4.3.3.1 Sampling plans for the following tests shall be as specified in Table I.

TABLE I

Test	Classification	Sample Size	Rejection	Test Procedure	Code No.
Spring embrittlement of tension spring (see 3.3)	Major	30	1	4.4.1	11006
Spring embrittlement of interrupter spring (see 3.3)	Critical	30	1	4.4.1	17008
Torque of setting sleeve on body assembly (see 3.4)	Major	40	1	4.4.2	20009
Support, firing pin load test (see dwg. 8798238)	Major	40	1	4.4.3	01010
Spring load of interrupter spring (see dwg. 8798231)	Major	30	1	4.4.5	17009

4.3.3.2 Body assembly interrupter non-arming, Critical defect, Code No. 20010 (see dwg. 8798222).-This test shall be performed 100 percent. Body assemblies found defective when tested as specified in 4.4.6 shall be rejected.

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4.3.3.3 Body assembly interrupter arming - Major defect, Code No. 20011 (see dwg. 8798222).-This test shall be performed 100 percent. Body assemblies found defective when tested as specified in 4.4.6 shall be rejected.

4.3.3.4 Body assembly interrupter arming retest, Code No. 20012.-At the option of the contractor, body assemblies found defective (see 4.3.3.3) shall be retested once as specified in 4.4.4.1. Body assemblies found defective when retested shall be rejected and not be returned to the lot. No retest is allowed for para 4.3.3.2, critical defect, body assembly interrupter non-arming.

4.3.3.5 Ogive Salt Spray Test.-Major defect, Code No. 20013 (see dwg. 8798220).-The sample plan and acceptance criteria for the test shall be as specified below. Test equipment shall be in accordance with 4.3.4. Any ogive which fails the test shall be cause for lot rejection. Ogives subjected to this test per 4.4.6 shall not be returned to the lot.

Test	Classification	Sample Size	Lot Acceptance Criteria	Test Procedure
a. First Article Salt Spray	Major	25	0 Failures	4.4.6
b. Production Salt Spray	Major	15	0 Failures	4.4.6

4.3.4 Inspection Equipment.-Index of Inspection Equipment List Number ET-9234453 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government

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all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test Methods and Procedures

4.4.1 Spring embrittlement.-Tension and interrupter springs shall be pulled to the minimum point of permanent distortion. Springs subjected to this test shall be scrapped.

4.4.2 Torque of setting sleeve on body assembly.-The torque required to rotate the setting sleeve shall be determined with equipment in accordance with 4.3.4. The body assembly shall be observed for compliance with the requirements.

4.4.3 Support, firing pin, load test.-The firing pin support shall be tested using equipment in accordance with 4.3.4. The initial required load shall be applied in an axial direction at a uniform rate of approximately 0.125 inch per minute. The final required load may be applied at an increased uniform rate of approximately 0.250 inch per minute. The load results shall be recorded to determine compliance with the requirements. Components subjected to this test shall be scrapped.

4.4.4 Body assembly interrupter arming and non-arming.-The unarmed assembly shall be subjected to this test by rotation in a Hurletron Fuze Body Inspection Machine (see 4.3.4). Each body assembly shall be placed in the Hurletron with the flash tube thread end facing up. Rotation will be first at the arming speed followed by rotation at the non-arming speed. Assemblies rejected by the Hurletron in the non-arming phase shall be classified defective and shall not be returned to the lot. Assemblies rejected by the Hurletron in the arming phase may be returned to the lot provided they successfully pass the retest specified in 4.4.4.1. These tests may be conducted before the setting sleeve assembly is staked to the body assembly.

4.4.4.1 Body assembly interrupter arming retest.-Body assemblies which fail the arming test described in 4.4.4 will be allowed one retest as described herein. The body assembly will be rotated in a suitable test fixture, with adequate illumination (see 4.3.4), other than the Hurletron machine. Rotation will be first at 2050 rpm; max., for arming, and then at 1450 rpm, min., for non-arming. The body assembly shall be placed in the fixture with the flash tube thread end facing up, and rotated in a counterclockwise direction when viewed from the top. The spindle shall be of the continuously operating type, or it shall start from rest for each test. The rate of acceleration imparted to each body assembly in the arming test shall be as nearly uniform as is practicable. An assembly will be

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classified defective if any part of the interrupter is visible within the flash hole during the arming test or if light is visible in the flash hole during the non-arming test. Assemblies found defective after this retest shall not be returned to the lot.

4.4.5 Spring, Interrupter Load Test.-The interrupter spring shall be tested using equipment in accordance with 4.3.4. The required loads shall be applied in an axial direction for a minimum of 24 hours. The load results shall be recorded to determine compliance with the requirements of dwg. 8798231 (Note 14).

4.4.6 Salt Spray.-The ogives shall be subjected to this test in accordance with Federal Test Method Standard No. 151, Method 811.1 for an exposure period of 96 hours using the 5 percent solution. After completion of the test, the components shall be observed for compliance with the requirements of the applicable plating specifications.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and Marking

5.1.1 Level C.-Packaging, packing and marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment design by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the

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contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit

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conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND7. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of first article samples.-Instructions as to the location for evaluation of the first article shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND, Dover, New Jersey 07801.

6.5 Supersession data.-This specification supersedes Military Specification MIL-F-60348A (MU).

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